

File Analysis Request

I see you uploaded files — but you didn't ask anything yet 😊

Tell me what you want me to do with them:

- Explain concepts (easy / exam style / deep understanding)
- Create notes / summaries
- Important questions
- Solve numericals
- Compare topics
- Prepare viva answers
- Make memory tricks
- Or full unit preparation

Just say the task and the unit 👍

JNTUV – AR23 (NLP Unit-I) — 2 Mark Answers

1. What are the origins of Natural Language Processing (NLP)? (2M)

Natural Language Processing originated from Artificial Intelligence research aimed at enabling computers to understand human language.

Historical phases:

- **1950s–1960s:** Rule-based symbolic systems (Turing Test, grammar rules)
 - **1970s–1980s:** Knowledge-based systems using semantics and logic
 - **1990s:** Statistical NLP using corpora and probability models
 - **2000s–Present:** Deep learning & Large Language Models (BERT, GPT)
-

2. Define a Language Model (LM). Why is it important in NLP? (2M)

A **Language Model** is a probabilistic model that assigns a probability to a sequence of words and predicts the next word in a sentence.

Example:

"I want to eat ____" → predicts *food/pizza*

Importance:

- Speech recognition
 - Machine translation
 - Text prediction & chatbots
 - Spelling correction
-

3. What is a Finite State Automaton (FSA)? (2M)

A **Finite State Automaton** is an abstract machine used to recognize patterns in input strings using states and transitions.

If the final state is accepting → string accepted; otherwise rejected.

4. Define tokenization with a suitable example. (2M)

Tokenization is the process of splitting a sentence into smaller units called tokens (words).

Example:

Sentence: "*NLP is interesting*"

Tokens → [NLP, is, interesting]

5. What is Minimum Edit Distance? Mention its applications. (2M)

Minimum Edit Distance is the minimum number of operations (insert, delete, substitute) required to convert one word into another.

Example:

cat → cut (substitution = 1)

Applications:

- Spell checking
 - Auto-correction
 - OCR correction
-

6. State the origins of NLP. (2M)

NLP evolved through four stages:

1. Rule-based systems (1950s)
 2. Knowledge-based semantic systems (1970s)
 3. Statistical machine learning models (1990s)
 4. Neural networks & deep learning (Modern era)
-

7. List any two challenges of NLP. (2M)

Any two:

- Ambiguity in language
 - Context understanding
 - Data sparsity
 - Multilingual processing
-

8. Define regular expression. (2M)

A **Regular Expression (Regex)** is a pattern of characters used to match or search text in strings.

Example:

`\d{3}` → matches any three-digit number

9. Define tokenization. (2M)

Tokenization is the process of dividing text into meaningful units such as words, numbers, or symbols called tokens.

10. Differentiate between grammar-based and statistical language models. (2M)

Grammar-Based LM	Statistical LM
Uses predefined grammar rules	Uses probability & corpus data

Grammar-Based LM	Statistical LM
Based on syntax structures (CFG)	Based on frequency of word sequences
Precise but rigid	Flexible and realistic
No learning capability	Learns from data

JNTUV – AR23 (NLP Unit-I) — 5 Mark Answers

1. Major challenges of Natural Language Processing with examples (5M)

Natural Language Processing is difficult because human language is complex, ambiguous, and context-dependent.

1. Ambiguity

A sentence can have multiple meanings.

Example:

"He saw the man with the telescope"

- He used a telescope
 - The man had a telescope
-

2. Context Understanding

Words change meaning depending on context.

Example:

bank → financial institution / river side

3. Morphological Variations

Words appear in many forms but share same root.

run → runs → running → ran

4. Syntax vs Semantics

Sentence can be grammatically correct but meaningless.

Example:

"Colourless green ideas sleep furiously."

5. World Knowledge (Common Sense)

Machines lack real-world understanding.

Example:

"John dropped the glass. It broke."

Human knows "it" = glass

6. Data Sparsity

Many valid word combinations never appear in training data.

7. Noisy Text

Slang and typos in social media.

Example: *u r gr8*

2. Grammar-based LM vs Statistical LM (5M)

Grammar-Based Language Model

Uses predefined grammar rules (CFG) to define sentence structure.

Example:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

Features

- Rule-based
 - No probability
 - Explains syntax clearly
 - Hard to scale
-

Statistical Language Model

Uses probability of word sequences learned from corpus.

Example:

$P(I \text{ want pizza})$

$= P(I) \times P(\text{want} | I) \times P(\text{pizza} | \text{want})$

Features

- Data-driven
 - Predicts next word
 - Handles real-world text
 - Needs large dataset
-

Differences

Grammar Based	Statistical
Rule based	Data driven
No learning	Learns from corpus
Handles syntax	Handles usage probability
Not scalable	Scalable

Grammar Based	Statistical
Cannot handle variation	Handles variation

3. English Morphology & Role of Finite State Transducers (5M)

English Morphology

Study of word formation and structure using **morphemes** (smallest meaning unit).

Types

1. Inflectional Morphology

Changes grammatical form but not meaning

walk → walks → walking → walked

2. Derivational Morphology

Creates new word and meaning

happy → happiness

play → player

Role of Finite State Transducers (FST)

FST maps input word → morphological output.

Functions

- Morphological analysis
 - Spell checking
 - Speech recognition
 - Machine translation
-

Example

running → run + ing

FST splits word into root and suffix.

4. Finite State Automata with NLP applications (5M)

Definition

Finite State Automaton (FSA) is a machine with states and transitions that accepts or rejects strings.

Components

- $Q \rightarrow$ states
 - $\Sigma \rightarrow$ input symbols
 - $q_0 \rightarrow$ initial state
 - $F \rightarrow$ final state
 - $\delta \rightarrow$ transition function
-

Types

- DFA — one transition
 - NFA — multiple transitions
-

Applications in NLP

1. Tokenization
 2. Lexical analysis
 3. Spell checking
 4. Pattern matching (regex)
 5. Speech processing
-
-

5. Spelling error detection and correction techniques (5M)

Spelling correction identifies incorrect words and suggests correct words.

1. Dictionary Lookup

Check word in vocabulary.

Example:

recieve → not found

2. Minimum Edit Distance

Choose closest valid word based on operations.

recieve → receive

3. N-gram Language Model

Choose word fitting context.

"I want to ___ pizza" → eat

4. Phonetic Matching

Words sounding similar corrected.

nite → night

5. Noisy Channel Model

Most probable correction based on probability.

6. Minimum Edit Distance with example (5M)

Definition

Minimum Edit Distance is the minimum number of edits needed to convert one string into another using:

- Insert
 - Delete
 - Substitute
-

Example

Convert **intention** → **execution**

Operations:

intention

→ inention (delete t)

→ enention (substitute i→e)

→ exention (substitute n→x)

→ execution

Applications

- Spell checking
 - OCR correction
 - DNA sequence matching
-

End of 5M Answers

JNTUV – AR23 (NLP Unit-I) — 10 Mark Answers

1. Explain Language Modeling in detail (10M)

A **Language Model (LM)** is a probabilistic model that assigns a probability to a sequence of words and predicts the next word in a sentence.

Example:

"I want to eat ____" → pizza / food

Mathematically:

$$P(w_1, w_2, \dots, w_n) = P(w_1)P(w_2 \mid w_1)P(w_3 \mid w_1, w_2) \dots P(w_n \mid w_1 \dots w_{n-1})$$

a) Grammar-Based Language Models

These models use predefined linguistic grammar rules (Context Free Grammar – CFG).

Example rules:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

Characteristics

- Rule-based
- No probability involved
- Syntax-oriented
- Explainable but rigid

Advantages

- Accurate grammatical structure
- Good for parsing and compilers

Disadvantages

- Cannot handle ambiguity
 - Hard to scale
 - Cannot learn from data
-

b) Statistical Language Models

Uses probability and corpus frequency to predict next word.

Chain Rule:

$$P(w_n \mid w_1 \dots w_{n-1})$$

Simplified using N-gram assumption:

Unigram: independent words

Bigram: depends on previous word

Trigram: depends on previous two words

Example:

$$P(I \text{ want pizza}) = P(I) \times P(\text{want} \mid I) \times P(\text{pizza} \mid \text{want})$$

Advantages

- Handles real language usage
- Learns from data
- Works with noisy text

Limitations

- Data sparsity
 - Needs large corpus
-

c) Applications of Language Models in NLP

1. Speech Recognition
 2. Machine Translation
 3. Chatbots
 4. Text prediction (keyboard suggestions)
 5. Spell correction
-
-

2. Tokenization and Spelling Error Detection (10M)

Tokenization

Tokenization is the process of splitting text into smaller units called tokens.

Example:

“NLP is powerful” → [NLP, is, powerful]

a) Types of Tokenization

1. Word Tokenization

Sentence → words

2. Sentence Tokenization

Paragraph → sentences

3. Subword Tokenization

play + ing

4. Character Tokenization

N L P

Spelling Errors

b) Common Spelling Errors

1. Insertion → comming
 2. Deletion → recieve
 3. Substitution → speling
 4. Transposition → teh
-

c) Detection & Correction Methods

Detection

- Dictionary lookup
- N-gram probability

Correction

1. Minimum Edit Distance
 2. Phonetic matching
 3. Noisy channel model
 4. Language model based correction
-
-

3. Minimum Edit Distance in detail (10M)

Minimum Edit Distance = minimum operations required to convert one word into another.

a) Edit Operations

- Insert
- Delete
- Substitute

Example:

cat → cut (1 substitution)

b) Dynamic Programming Approach

Create matrix of size $(m+1) \times (n+1)$

Steps:

1. Initialize first row/column
2. Compute cost using recurrence:

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 \\ D(i, j-1) + 1 \\ D(i-1, j-1) + \text{cost} \end{cases}$$

3. Final cell gives minimum edits
-

c) Applications

- Spell correction
 - OCR correction
 - Speech recognition
 - DNA sequence matching
-
-

4. Regular Expressions and Finite State Automata (10M)

Regular Expressions

Pattern matching strings using symbols.

Example:

$\backslash d\{3\}$ → matches 3 digit numbers

Uses

- Tokenization
 - Email extraction
 - URL detection
-

Finite State Automata (FSA)

Abstract machine recognizing strings using states and transitions.

Components

Q – states

Σ – symbols

q_0 – start state

F – final state

δ – transition

Types

- DFA
 - NFA
-

Example

Accept strings ending with 'a'.

Applications

- Lexical analysis
 - Pattern matching
 - Spell checking
 - Speech processing
-
-

5. Tokenization, Spelling Detection and Correction (10M)

Tokenization

Splitting text into words.

Spelling Error Detection

Identify invalid words using dictionary.

Example:

recieve → incorrect

Spelling Correction Techniques

1. Minimum Edit Distance
 2. Phonetic matching
 3. Language model probability
 4. Noisy channel model
-
-

6. English Morphology & Finite State Transducers (10M)

English Morphology

Study of word structure using morphemes.

Types

Inflectional

walk → walking

Derivational

happy → happiness

Finite State Transducers (FST)

Maps input word → structured output.

Example:

running → run + ing

Role

- Morphological analysis
 - Spell checking
 - Machine translation
 - Speech recognition
-

End of 10M Answers

JNTUV – AR23 (NLP Unit-II) — 2 Mark Answers

1. What is an unsmoothed N-gram model? (2M)

An **unsmoothed N-gram model** estimates the probability of a word sequence using only observed frequency counts from the corpus (Maximum Likelihood Estimation). Unseen word combinations receive probability **zero**.

2. Define smoothing in N-gram language models (2M)

Smoothing is the process of redistributing probability mass from frequent N-grams to unseen N-grams so that no valid word sequence gets zero probability.

3. What are word classes? Give one example (2M)

Word classes are grammatical categories grouping words with similar syntactic properties.

Example:

Noun – *computer*

4. Define Part-of-Speech (PoS) tagging (2M)

PoS tagging is the process of assigning a grammatical tag (noun, verb, adjective, etc.) to each word in a sentence based on its context.

5. What is a Hidden Markov Model (HMM)? (2M)

A **Hidden Markov Model** is a probabilistic sequence model where hidden states generate observable outputs, commonly used for PoS tagging and sequence prediction.

6. What is an N-gram? (2M)

An **N-gram** is a contiguous sequence of N words from a text.

Example:

Bigram → "natural language"

7. What is data sparsity in language modeling? (2M)

Data sparsity occurs when many valid word combinations never appear in the training corpus, resulting in zero probability estimates.

8. Define interpolation and backoff models with examples (2M)

Backoff: Use higher-order N-gram if available; otherwise use lower-order.

Example: trigram → bigram → unigram

Interpolation: Combine probabilities from unigram, bigram, and trigram simultaneously.

9. What is a Maximum Entropy model? (2M)

A **Maximum Entropy model** is a probabilistic classification model that predicts outcomes using contextual features without assuming independence between them.

10. Explain unsmoothed N-gram language models (2M)

In unsmoothed N-gram models, probability is computed using relative frequency counts:

$$P(w_n | w_{n-1}) = \frac{\text{Count}(w_{n-1}, w_n)}{\text{Count}(w_{n-1})}$$

They fail for unseen sequences because probability becomes zero.

JNTUV – AR23 (NLP Unit-II) — 5 Mark Answers

1. Evaluation methods for N-gram models (5M)

Evaluation determines how well a language model predicts real language.

1. Extrinsic Evaluation (In-vivo evaluation)

Model is tested inside an application.

Examples

- Speech recognition accuracy
- Machine translation quality

If application performance improves → model is good.

But expensive and time-consuming.

2. Intrinsic Evaluation

Measures model quality directly using mathematical metric.

Perplexity (Most important metric)

Perplexity measures how well the model predicts a test sequence.

$$PP(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}}$$

Lower perplexity → better model

Observation

Higher order N-gram → lower perplexity

Model	Perplexity
Unigram	High
Bigram	Medium
Trigram	Low

2. Smoothing techniques in N-gram language models (5M)

Problem: unseen N-grams get probability 0 (data sparsity).

Smoothing redistributes probability mass.

1. Laplace (Add-One) Smoothing

Add 1 to every count

$$P(w_i) = \frac{c_i + 1}{N + V}$$

Removes zero probability

2. Add-k Smoothing

Add small fractional value k instead of 1

Better than Laplace

3. Backoff

Use lower-order N-gram if higher order missing

trigram → bigram → unigram

4. Interpolation

Combine all N-gram probabilities

$$P = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

5. Kneser-Ney Smoothing

Advanced method using continuation probability

3. Rule-based, Stochastic and Transformation-based POS tagging (5M)

POS tagging assigns grammatical tags to words.

1. Rule-Based Tagging

Uses dictionary + hand-written grammar rules.

Example rule:

If word between determiner and noun → adjective

Advantages

- Linguistically accurate

Disadvantages

- Hard to build rules
 - Not scalable
-

2. Stochastic Tagging

Uses probability from corpus.

Methods:

- Word frequency tagging
- Tag sequence probability

Chooses most frequent tag

3. Transformation-Based Tagging (TBL)

Combination of rule-based + statistical

Steps:

1. Assign initial tag
 2. Apply learned transformation rules
 3. Correct tags iteratively
-
-

4. Interpolation vs Backoff models (5M)

Feature	Backoff	Interpolation
Use lower order	Only if higher missing	Always
Probability	Single model at a time	Combined models
Robustness	Context dependent	Smoother
Performance	Good	Better generally

Backoff

$$P(w_n \mid w_{n-2}, w_{n-1}) \rightarrow \text{ifzero} \rightarrow P(w_n \mid w_{n-1})$$

Interpolation

$$P = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

5. Word classes with examples (5M)

Word classes = grammatical categories grouping words with similar properties.

Open Class (Content Words)

New words can be added.

Class	Example
Noun	computer
Verb	run
Adjective	happy
Adverb	quickly

Closed Class (Function Words)

Fixed vocabulary

Class	Example
Determiner	the
Pronoun	he
Preposition	on
Conjunction	and

6. Working of Hidden Markov Model for POS tagging (5M)

HMM is probabilistic sequence model with hidden states (tags) and observed words.

Concept

- Words → observations
- POS tags → hidden states

We find tag sequence maximizing probability:

$$P(W, T) = P(T)P(W | T)$$

Steps

1. Calculate transition probability
 $P(\text{tag}_i | \text{tag}_{i-1})$
 2. Calculate emission probability
 $P(\text{word} | \text{tag})$
 3. Use Viterbi algorithm to find best tag sequence
-

Example

Sentence:

Time flies fast

Possible tags computed → most probable sequence chosen

Applications

- POS tagging
 - Speech recognition
 - Named Entity Recognition
-

End of 5M Answers (Unit-II)

JNTUV – AR23 (NLP Unit-II) — 10 Mark Detailed Answers

1. Part-of-Speech Tagging Approaches (10M)

Part-of-Speech (PoS) tagging is the process of assigning a grammatical category (noun, verb, adjective etc.) to each word in a sentence based on context.

Example:

“Book the flight” → *Book* = *Verb*

“Read the book” → *Book* = *Noun*

A) Rule-Based Tagging

Uses dictionary + manually written linguistic rules.

Working

1. Find all possible tags from lexicon
2. Apply context rules to remove wrong tags

Example Rule

If word is preceded by determiner and followed by noun → adjective

“A nice car” → *nice* = *adjective*

Advantages

- Linguistically accurate
- Human interpretable

Limitations

- Needs many rules (~1000)
 - Hard to scale
 - Time-consuming
-

B) Stochastic (Probabilistic) Tagging

Uses probabilities from tagged corpus.

Methods

1. Word Frequency Method

Choose most frequent tag

play → usually Verb

2. Tag Sequence Probability

Uses previous tags

If previous tags: N V D

Next tag probability highest → Verb

Advantages

- Learns automatically
- Works well with large corpus

Disadvantages

- Needs training data
 - Unknown words problem
-

C) Transformation-Based Tagging (Brill Tagger)

Combination of rule-based + statistical.

Working Steps

1. Assign initial tags (most frequent)
2. Learn transformation rules
3. Apply corrections iteratively

Example Rule:

Change noun → verb if preceded by "to"

Advantages

- High accuracy (~97%)
- Interpretable rules

Limitations

- Training time high
-

2. Word-Level Analysis in NLP (10M)

Word-level analysis is the first stage of NLP focusing on individual words.

Tasks Involved

1. Identifying words
 2. Understanding form
 3. Understanding meaning
 4. Handling word variations
-

1. N-gram Language Modeling

Predict next word using previous words.

$$P(w \mid h)$$

Example:

“its water is so transparent that ___” → the

2. Word Classes

Words grouped into grammatical categories:

- Noun
 - Verb
 - Adjective
 - Adverb
-

3. PoS Tagging

Assign tag to each word

Input: words → Output: tags

4. Morphological Processing

Handles word forms

run → running → ran

5. Hidden Markov & Maximum Entropy Models

Used for sequence prediction in tagging

Importance

- Foundation for syntax & semantics
 - Improves machine translation
 - Improves speech recognition
-
-

3. Maximum Entropy Model for POS Tagging (10M)

Maximum Entropy (MaxEnt) is a probabilistic classification model that uses contextual features without independence assumptions.

Idea

Choose tag with highest probability given features:

$$P(\text{tag} \mid \text{context})$$

Model Structure

Uses features such as:

- Current word
 - Previous word
 - Capitalization
 - Suffix/prefix
-

Working

1. Extract features from sentence
2. Compute probability distribution
3. Select tag with maximum probability

Example

“running”

suffix = ing → Verb likely

Advantages

- High accuracy
- Handles multiple features
- Flexible

Limitations

- Computationally expensive
 - Needs large training data
-
-

4. N-gram Language Models (10M)

N-gram predicts next word using previous N-1 words.

a) Unsmoothed N-grams

Probability from counts:

$$P(w_n \mid w_{n-1}) = \frac{\text{Count}(w_{n-1}, w_n)}{\text{Count}(w_{n-1})}$$

Problem: zero probability

b) Smoothing Techniques

Redistribute probability mass.

1. Laplace (Add-one)
 2. Add-k
 3. Kneser-Ney
-

c) Interpolation & Backoff

Backoff:

Use lower order if higher missing

Interpolation:

Combine all probabilities

$$P = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

5. Part-of-Speech Tagging in Detail (10M)

a) Word Classes

Open class:

- noun, verb, adjective, adverb

Closed class:

- pronoun, preposition, determiner
-

b) Rule-Based Tagging

Dictionary + grammar rules

c) Issues in PoS Tagging

1. Ambiguity
"back" → noun/verb/adverb
 2. Unknown words
 3. Context dependence
 4. Multi-tag words
-
-

6. Hidden Markov Models & Maximum Entropy Models (10M)

Hidden Markov Model

Structure

- Hidden states $\rightarrow$ POS tags
 - Observations $\rightarrow$ words
-

Working

Compute:

Transition probability

$$P(tag_i \mid tag_{i-1})$$

Emission probability

$$P(word \mid tag)$$

Use Viterbi algorithm to find best sequence.

Advantages

- Works well sequential data
- Efficient

Limitations

- Independence assumption
 - Limited features
-

Maximum Entropy Model

Structure

Feature-based classifier

$$P(tag \mid context)$$

Advantages

- Uses many features

- Higher accuracy

Limitations

- Slow training
 - Complex computation
-

End of Unit-II 10M Answers